DATA ITEM DESCRIPTION

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1. TITLE	2. IDENTIFICATION NUMBER
SOFTWARE CERTIFICATION PLAN (SCP)	DI-NUOR-81412

3. DESCRIPTION/PURPOSE

3.1 This plan defines the nuclear certification process and the process or activities that are necessary to design in the characteristics of nuclear safe software from the start of the development cycle throughout the system's lifecycle.

4. APPROVAL DATE (YYMMDD)	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC APPLICABLE	6b. GIDEP APPLICABLE
940729	F/SA-ALC-NWI		

7. APPLICATION/INTERRELATIONSHIP

7.1 This Data Item Description (DID) contains the format and content preparation instructions for data resulting from the work task described by 90.11 of MIL-STD-1822.

7.2 This plan is related to DI-NUOR-81409 , Nuclear Certification Plan where the top level (tier one) tasks are defined. The lower level tasks are incorporated in the SCP.

8. APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		F7049

10. PREPARATION INSTRUCTIONS

10.1 Format. Contractor format is acceptable.

10.2 Content. The content of the SCP shall be as defined below:

10.2.1 The plan shall focus on the process the software will go through to achieve nuclear certification, as an integral part of the system's nuclear certification. The plan shall define how to ensure that the system software design process incorporates nuclear safety requirements.

10.2.2 The plan shall define each step (and the task that is associated with that nuclear certification process step) in the certification process, what software or documentation products are to be produced, how these products are to be evaluated, and explicit pass or fail criteria to ensure that the nuclear requirements defined in the requirements analysis are not diluted or corrupted in the final software product.

10.2.3 The plan shall define how the applicable nuclear safety criteria is integrated in the software design to include:

a. Software requirements analysis

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11. DISTRIBUTION STATEMENT

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

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Block 10, Preparation Instructions (Continued)

b. Preliminary software design

c. Detailed software design

d. Coding and unit testing

e. Computer software component integration

f. Computer software configuration item testing

g. Configuration management

h. Software quality evaluation

10.2.4 The plan shall specifically state the steps to ensure that:

a. Each nuclear safety requirement is traceable to one or more critical nuclear functional unit or module.

b. Critical nuclear functions or data is analyzable or testable to ensure proper implementation is accomplished.

c. Access to the critical nuclear functions or data is controlled.

d. Each critical nuclear function is divided into several distinct and sequential steps (for aircraft systems only).

e. The avionics control software is the only way to access the critical nuclear functions; no other access is possible (for aircraft systems only).

f. Verification of intent to control and monitor nuclear weapons where: (1) nuclear critical function software is independent of the processor memory used in normal avionic system function, and (2) access to the nuclear critical memory is tightly controlled (for aircraft systems only).

10.2.5 The plan shall contain the software development milestones, task event timelines, a definition of the deliverable software and software documentation products, and the system and nuclear certification milestones.

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